Multiple Linear Regression: Endurance Performance

Background

Endurance race performance depends on physiological capacity, training volume, and body characteristics. This analysis uses multiple linear regression to model 10 km race times for recreational and competitive runners, with modeling decisions and results interpreted in context. Graphical diagnostics are used to support assumptions and interpretation.

Executive Summary

This study investigates the factors influencing 10 km race performance in recreational and competitive runners using multiple linear regression. VO_2max and weekly mileage are the strongest predictors, with higher values associated with faster race time. The refined model, after removing invalid and influential data points, explains about 51% of the variance and meets key regression assumptions (Adjusted R-squared: 0.51) . Limitations include missing factors such as training intensity, gender, and injury history, and the results reflect associations rather than causal effects.

1. Exploratory Analysis

1.1 Dataset Description

Response Variable: RaceTime – 10 km race time (minutes)

Predictors:

- VO_2max – Estimated VO_2max ($ml \cdot kg^{-1} \cdot min^{-1}$)
- WeeklyMileage – Average weekly running distance (km)
- BodyMass – Body mass (kg)
- Age – Age (years)
- TrainingYears – Years of structured endurance training

Sample size: ~300 observations

1.2 Summary Statistics

Table1: Summary statistics for all variables in the endurance dataset

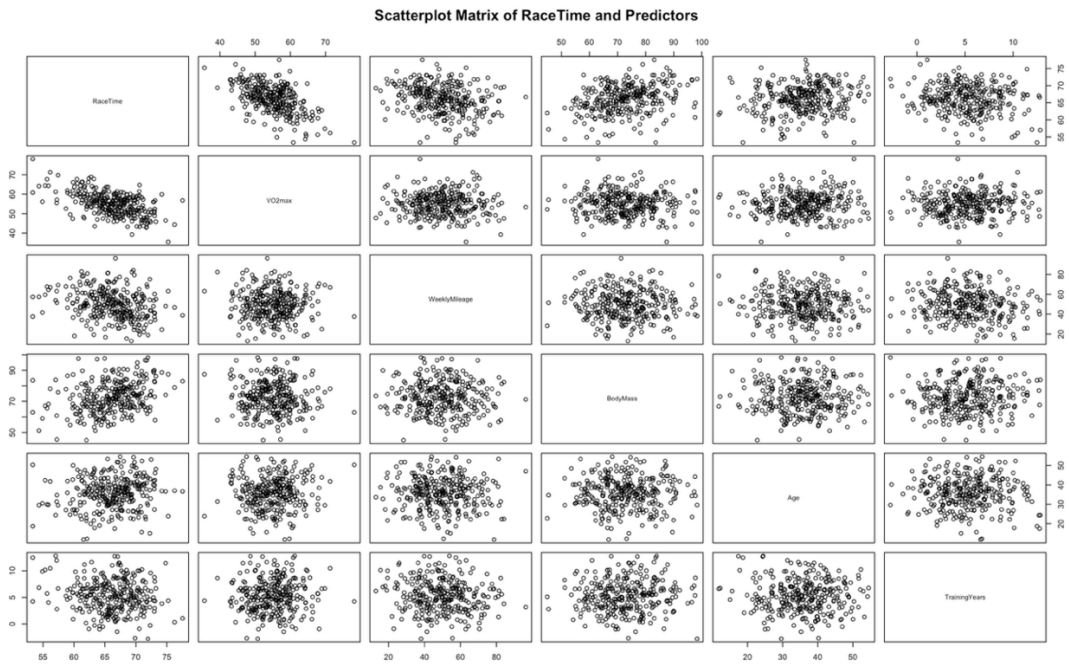
Variable	Min	1st Qu.	Median	Mean	3rd Qu.	Max	SD
RaceTime	53.37	63.46	66.47	66.16	69.15	77.55	4.17
VO_2max	35.55	50.9	55.36	54.97	58.76	78.12	5.91
WeeklyMileage	12.93	39.43	49.72	49.68	59.25	96.18	14.42
BodyMass	45.03	66.29	72.43	72.82	79.11	98.32	9.97
Age	11.83	30.55	36.32	35.79	41	54.52	8.08
TrainingYears	-2.77	3.17	5.17	5.28	7.13	12.81	2.99

Observations:

RaceTime (66.16 ± 4.17 min) shows moderate variability. VO_2max (54.97 ± 5.91 mL/kg/min) and WeeklyMileage (49.68 ± 14.42 km) also vary across participants, with WeeklyMileage being the most dispersed. BodyMass (72.82 ± 9.97 kg) and Age (35.79 ± 8.08 years) are approximately symmetric. TrainingYears includes a negative value (-2.77), indicating a likely data entry error. Overall, the dataset provides adequate variability for regression analysis with no other extreme outliers.

1.3 Scatter Plots

Figures 1: RaceTime vs each predictor



As shown in Figure 1, VO_2max exhibits the strongest negative association with RaceTime, indicating that higher aerobic capacity is linked to faster race performance. WeeklyMileage shows a moderate negative relationship, while BodyMass displays a weak positive association. Age and TrainingYears show little direct relationship with RaceTime. VO_2max and WeeklyMileage are moderately correlated, suggesting potential multicollinearity that warrants further assessment in the regression analysis.

2. Full Model Specification

2.1 Full Model

- VO_2max and WeeklyMileage have negative coefficients: higher aerobic capacity and greater weekly mileage are associated with faster race times.
- BodyMass and Age have positive coefficients: higher weight and older age are associated with slightly slower race times.
- TrainingYears shows a small negative effect.
- All predictors are statistically significant at the 5% level.
- The model explains approximately 50% of the variance (Adjusted $R^2 \approx 0.5$).

2.2 Model Fitting

The fitted regression equation is:

$$RaceTime = 81.31 - 0.39VO_2max - 0.08WeeklyMileage + 0.12BodyMass + 0.08\ Age - 0.22TrainingYears$$

2.3 Coefficient Interpretation

The coefficient for VO_2max is −0.39. This means that, holding all other predictors constant, every 1 $ml \cdot kg^{-1} \cdot min^{-1}$ increase in VO_2max is associated with an average decrease of 0.39 minutes in 10 km race time (approximately 23 seconds).

In practical terms, athletes with higher maximal oxygen uptake tend to complete 10 km races faster, even after accounting for weekly mileage, body mass, age, and training experience. (Note: This is an association, not a causal effect, because this is observational data.)

3. Inference and Model Fit

3.1 Significant Predictors

VO_2max, WeeklyMileage, BodyMass, Age, and TrainingYears are statistically significant predictors of RaceTime at the 5% level ($p < 0.001$).

3.2 Meaning of Statistical Significance

A significant p-value indicates evidence of a linear association between the predictor and RaceTime, controlling for the other variables. This reflects correlation, not causation.

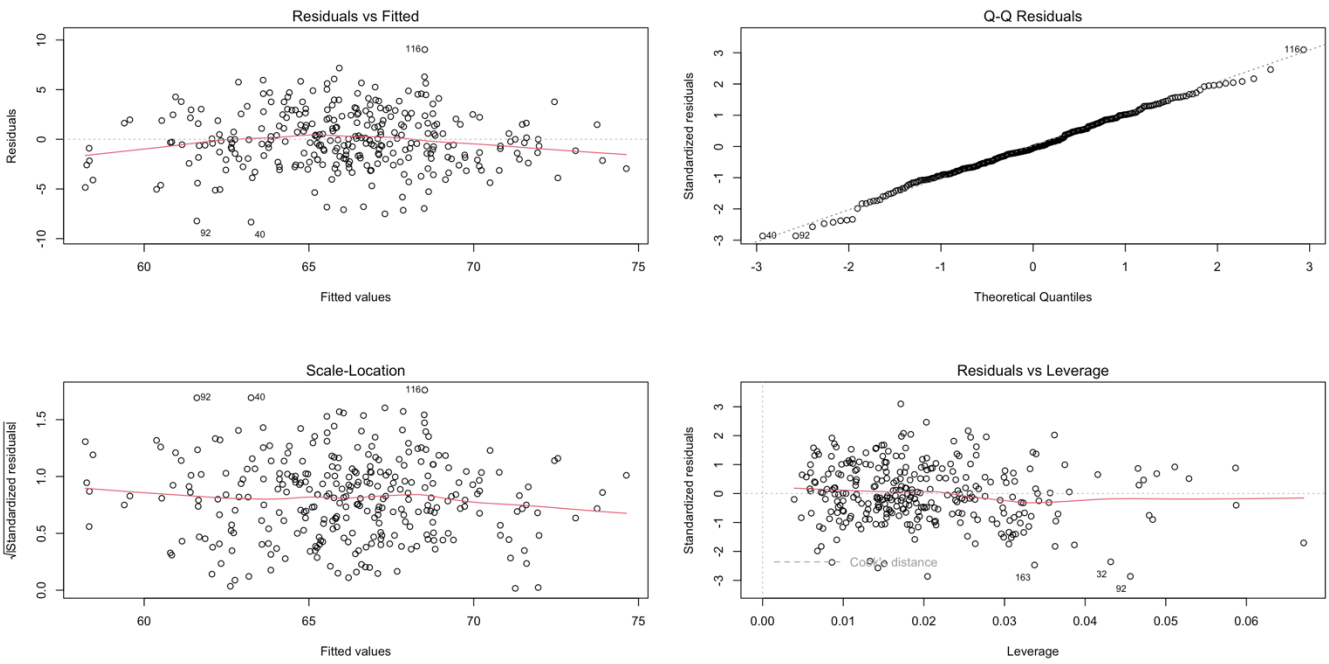
3.3 Model Fit

Adjusted $R^2 = 0.5025$, indicating that the model explains approximately 50% of the variance in RaceTime.

4. Model Diagnostics

4.1 Residual Analysis

Figure 2: Residual Diagnostics for the Multiple Linear Regression Model



As shown in Figure 2 (Residuals vs Fitted), residuals are randomly scattered around zero, indicating no major violations of linearity or homoscedasticity.

4.2 Normality of Residuals

Normality of residuals was assessed using a Q–Q residuals plot (Figure2) and the Shapiro–Wilk test. The Q–Q plot shows that residuals lie approximately along the reference line, and the Shapiro–Wilk test does not reject the null hypothesis of normality ($W = 0.995$, $p = 0.502$). Therefore, the normality assumption appears to be reasonably satisfied.

4.3 Multicollinearity

There is no evidence of multicollinearity among the predictors, as all VIF values are close to 1.

4.4 Overall Assumptions

Based on the diagnostic plots, the assumptions of linearity, normality of residuals, constant variance, and absence of major influential observations appear to be reasonably satisfied. The multiple linear regression model is therefore appropriate for inference.

5. Model Refinement

5.1 Reduced Model

After removing negative TrainingYears and high-leverage points, the refined regression model is:

$$RaceTime = 81.12 - 0.39VO_2max - 0.08WeeklyMileage + 0.12BodyMass + 0.08\ Age - 0.24TrainingYears$$

All predictors are retained because each contributes meaningfully to predicting RaceTime.

5.2 Justification for Exclusion

- No predictors were removed by AIC-based stepwise selection because all variables improve model fit.
- The removal of extreme or invalid data (TrainingYears < 0 and high-leverage points) ensures more reliable coefficient estimates without changing the overall structure of the model.

5.3 Comparison of Reduced Model to Full Model:

- Coefficients: Largely similar; removing negative TrainingYears or influential points slightly changes some estimates (TrainingYears from -0.22 to -0.24).
- Adjusted R^2: Stable (full: 0.502, no negatives: 0.497, no influential: 0.510).
- AIC: Lower for cleaned models (full: 1506.65, no negatives: 1461.95, no influential: 1416.51), indicating better fit.
- Conclusion: The model after removing influential points is preferred—similar predictive power but more robust and better fitting.

6. Interpretation and Limitations

6.1 Non-causal Predictor

WeeklyMileage coefficient should not be interpreted causally. Although higher mileage is associated with faster race times, it may be confounded by VO_2max, age, or training experience. Observational data do not allow causal inference.

6.2 Dataset Limitation

- Important factors such as training intensity, gender, or injury history are not included, limiting predictive accuracy.
- Extreme or unrealistic values can also bias predictions if not cleaned.

(All R code used for data cleaning, analysis, and model diagnostics is provided in a separate appendix file*, Appendix_Endurance_Multiple_Linear_Regression.R*.)